INFINITELY MANY INTEGERS THAT ARE NOT QUOTIENTS OF ZERO-FREE BALANCED TERNARY NUMBERS

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ABSTRACT. Let $\mathcal{B}$ be the set of positive integers whose balanced ternary representation uses only the digits 1 and -1. Selfridge and Lacampagne asked whether every integer $n \not\equiv 0 \pmod 3$ is a quotient of two elements of $\mathcal{B}$. Bai, Meleshko, Riasat and Shallit found seventeen exceptions, the largest 3649, and remarked that there are likely infinitely many, without proof. We prove that $4\cdot 3^k+5$, $4\cdot 3^k-5$, $8\cdot 3^k+7$ and $8\cdot 3^k+17$ lie outside $\mathcal{B}/\mathcal{B}$ for every $k \geq 5$, and $8\cdot 3^k-7$ for every $k \geq 7$; each bound on k is sharp. The proof exhibits, for each family, a set of carries of size linear in k that is closed under the carry map of multiplication by n and contains no accepting carry. For $4\cdot 3^k+5$ the argument is short enough to check by hand. The same method, applied to one residue class of k modulo 2 or 4 at a time, proves eight further families, among them $5\cdot 3^k+4$ for $k \equiv 0 \pmod 4$ and $5\cdot 3^k-4$ for $k \equiv 2 \pmod 4$. All thirteen results are formally verified in Lean 4, apart from the sharpness of the bounds, which rests on explicit witnesses. Of the 200 exceptions below 10^8, 75 are covered. For the analogous problem with the digits 0 and 1 we confirm one further case of the guess of Bai et al. that $621\cdot 3^{4k}-20$ always gives an exception.

Status of this preprint. The thirteen non-membership statements of Theorems 1.1 and 1.3 are formally verified in Lean 4 (Section 5), using only Lean's standard axioms. The sharpness clauses of both theorems and Proposition 6.1 are not in Lean: they are checked by a separate program (see *Code and data*), which recomputes the witnesses and the carry search, together with the other finite claims listed there. This version has not yet been independently reviewed.

1. INTRODUCTION

Can every integer that is not divisible by 3 be written as a quotient N/m of two positive integers whose balanced ternary digits are all 1 or -1? For example, $8 = 9-1$ has a zero digit, but $8 = 16/2$ with $16 = 27-9-3+1$ and $2 = 3-1$. John Selfridge and Carole Lacampagne asked this question, and Guy records it in Section F31 of *Unsolved Problems in Number Theory* [3, pp. 403–404], noting that experiments suggested the answer yes. Bai et al. trace it to §7 of the paper of Loxton and van der Poorten on the analogous problem in base four [4]. Questions of this kind ask which integers lie in the quotient set of a set defined by its digits; Rowland and Shallit place such quotient sets in the framework of automatic sets of rational numbers and list related decision problems among their open questions [5].

To fix notation, every integer x has a unique balanced ternary representation $x = \sum_{j=0}^{L-1} \varepsilon_j 3^j$ with $\varepsilon_j \in \{-1, 0, 1\}$ and, if $x \neq 0$, $\varepsilon_{L-1} \neq 0$. Call x *zero-free* if $x \neq 0$ and every ε_j is nonzero, and let

$$\mathcal{B} = \{1, 2, 4, 5, 7, 11, 13, 14, 16, 20, 22, \dots\}$$

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be the set of positive zero-free integers. Since every element of $\mathcal{B}$ is prime to 3, so is every integer in the quotient set $\mathcal{B}/\mathcal{B} = \{N/m : N, m \in \mathcal{B}\}$, and the question is whether every integer $n \equiv \pm 1 \pmod 3$ lies in $\mathcal{B}/\mathcal{B}$.

The answer is no. Bai, Meleshko, Riasat and Shallit [1] found the exceptions

$$247,\ 277,\ 967,\ 977,\ 1211,\ 1219,\ 1895,\ 1937,\ 1951,$$
$$1961,\ 2183,\ 2191,\ 2911,\ 2921,\ 3029,\ 3641,\ 3649,$$

the first of which had also been found by Coppersmith [3, F31], and wrote: "It seems likely that there are infinitely many such counterexamples, but we have no proof." We supply a proof.

Theorem 1.1. *For every $k \geq 5$, none of*

$$4 \cdot 3^k + 5, \qquad 4 \cdot 3^k - 5, \qquad 8 \cdot 3^k + 7, \qquad 8 \cdot 3^k + 17$$

lies in $\mathcal{B}/\mathcal{B}$, and for every $k \geq 7$, $8 \cdot 3^k - 7$ does not lie in $\mathcal{B}/\mathcal{B}$. The bounds on k are sharp: the four numbers with $k = 4$, and $8 \cdot 3^6 - 7$, all lie in $\mathcal{B}/\mathcal{B}$.

Corollary 1.2. *There are infinitely many positive integers $n \not\equiv 0 \pmod 3$ with $n \notin \mathcal{B}/\mathcal{B}$.*

Several further families are exceptions only along a residue class of k, or are proved here only along one.

Theorem 1.3. *The following integers are not in $\mathcal{B}/\mathcal{B}$:*

1. *$8 \cdot 3^k - 17$ and $16 \cdot 3^k + 17$ for every even $k \geq 6$;*
2. *$10 \cdot 3^k - 11$, $10 \cdot 3^k - 19$, $20 \cdot 3^k - 19$ and $5 \cdot 3^k - 4$ for every $k \equiv 2 \pmod 4$ with $k \geq 6$;*
3. *$5 \cdot 3^k + 4$ for every $k \equiv 0 \pmod 4$ with $k \geq 8$, and $40 \cdot 3^k + 41$ for every $k \equiv 0 \pmod 4$ with $k \geq 12$.*

Each bound is sharp within its residue class: the previous member of the class lies in $\mathcal{B}/\mathcal{B}$.

For the first five families in Theorem 1.3, every exception below 10^8 of the form $a \cdot 3^k + b$ lies in the stated residue class, so the theorem accounts for all of them in that range.

Six of the seventeen exceptions above, namely 967, 977, 1951, 1961, 2911 and 2921, belong to our families. Of the 200 exceptions below 10^8 (Section 7), 50 belong to the five families of Theorem 1.1 and 75 to the families of Theorems 1.1 and 1.3 together.

The method is a finite automaton argument made uniform in k. Multiplying a zero-free number by n from the least significant digit upwards is governed by a carry, and $n \in \mathcal{B}/\mathcal{B}$ exactly when some sequence of carries starting at 0 ends in a carry that is itself 0 or positive and zero-free. For $n = a \cdot 3^k + b$ the carries that can occur form a set S_k whose elements have the shape $\pm(gX + \beta - t \cdot 3^i)$ with $X = 3^k$, a handful of fixed rationals g and integers β, t, and an index i running over a range that grows with k. Once S_k is written down, closure under the carry map is a finite list of identities between polynomials in X and 3^i, and the absence of an accepting carry follows from a gap in the balanced ternary digits of each candidate. Section 2 sets up the carry automaton and states the criterion we use. Section 3 proves the case $4 \cdot 3^k + 5$ in full. Section 4 treats the remaining families, whose carry sets are larger, and Section 5 describes the formal verification. Section 6 reports one computation on the companion problem for the digits 0 and 1, and Section 7 collects data and open questions.

2. The carry automaton

For an integer v let $\langle v \rangle \in \{-1, 0, 1\}$ denote the residue with $v \equiv \langle v \rangle \pmod 3$. The lowest balanced ternary digit of x is $\langle x \rangle$, and uniqueness of the representation gives the following peeling rule.

Lemma 2.1. *Let $x \neq 0$ and $y = (x - \langle x \rangle)/3$. Then x is zero-free if and only if $\langle x \rangle \neq 0$ and either $y = 0$ or y is zero-free.*

Fix $n \geq 2$. For a carry $r \in \mathbb{Z}$ and a digit $d \in \{1, -1\}$, put $v = r + nd$. If $\langle v \rangle \neq 0$ we say that the *move* (r, d) produces the digit $\langle v \rangle$ and leads to the carry $r' = (v - \langle v \rangle)/3$; if $\langle v \rangle = 0$ the move is *dead*. For instance, with $n = 5$ the move $(0, 1)$ gives $v = 5 \equiv -1$, so it produces the digit -1 and leads to the carry 2; since $2 = 3 - 1$ is zero-free, this one step already exhibits $5 \cdot 1 = 5 = (1\bar{1}\bar{1})_3$ (writing $\bar{1}$ for -1), and $5 \in \mathcal{B}/\mathcal{B}$. The next proposition is the only link between carries and the set $\mathcal{B}/\mathcal{B}$ that we need.

Proposition 2.2. *Let $n \geq 2$ and let $S \subseteq \mathbb{Z}$ satisfy*

(i) $0 \in S$;
(ii) *every move (r, d) with $r \in S$ that is not dead leads to a carry in S;*
(iii) $r \geq 2 - n$ *for every $r \in S$;*
(iv) *if $r \in S$ and the move $(r, 1)$ is not dead and leads to $r' > 0$, then r' is not zero-free.*

Then $n \notin \mathcal{B}/\mathcal{B}$.

Proof. Suppose $N = nm$ with $m, N \in \mathcal{B}$, and write $m = \sum_{j=0}^{L-1} d_j 3^j$ with $d_j \in \{1, -1\}$; since $m > 0$, the leading digit is $d_{L-1} = 1$. Define carries by $r_0 = 0$ and $r_{j+1} = (r_j + nd_j - e_j)/3$ with $e_j = \langle r_j + nd_j \rangle$, and put $m_J = \sum_{j \geq J} d_j 3^{j-J}$ and $M_J = r_J + nm_J$, so that $M_0 = N$. Every m_J with $J < L$ is a positive zero-free number, so $m_J \geq 1$.

We show by induction on $J \leq L - 1$ that $r_J \in S$ and M_J is zero-free. For $J = 0$ this is (i) and the hypothesis on N. Suppose it holds for some $J < L - 1$. Since $M_J = r_J + nd_J + 3nm_{J+1}$, we have $\langle M_J \rangle = e_J$, and Lemma 2.1 shows that $e_J \neq 0$ and that $M_{J+1} = (M_J - e_J)/3$ is either 0 or zero-free. Thus the move (r_J, d_J) is not dead, and $r_{J+1} \in S$ by (ii). By (iii), $M_{J+1} = r_{J+1} + nm_{J+1} \geq 2 - n + n > 0$, so M_{J+1} is zero-free.

Apply the same step at $J = L - 1$, where $m_{L-1} = d_{L-1} = 1$ and $M_L = r_L$. The move $(r_{L-1}, 1)$ is not dead, $r_L = (r_{L-1} + n - e_{L-1})/3 \geq (2 - n + n - 1)/3 > 0$, and $M_L = r_L$ is zero-free because it is nonzero. This contradicts (iv). □

Conversely, if S is the set of carries reachable from 0 and some reachable move $(r, 1)$ leads to 0 or to a positive zero-free carry, then reading the digits along the path gives m and $N = nm$ in $\mathcal{B}$. So the reachable set decides membership, and a computer search over it reproduces exactly the seventeen exceptions of [1] below 3650. For the families of Theorem 1.1 the reachable set is the set S_k we describe (we checked equality for $10 \leq k \leq 24$), but the proofs only use conditions (i) to (iv).

The map $(r, d) \mapsto (-r, -d)$ negates v, the digit and the new carry. Hence, if $S = -S$, it suffices to verify (ii) for the elements $r \geq 0$ of S and both digits.

3. The family $4 \cdot 3^k + 5$

Let $k \geq 5$, $X = 3^k$, $n = 4X + 5$ and $c = (n - 1)/2 = 2X + 2$. Define

$$S_k = \{0\} \cup \pm\{\, c,\ c - 2 \cdot 3^i,\ c - 4 \cdot 3^i \ :\ 0 \leq i \leq k - 1 \,\},$$

a set of $4k + 3$ integers. We verify the four conditions of Proposition 2.2.

Closure. Since $3 \mid X$, and $3 \mid 3^i$ when $i \geq 1$, each residue $\langle r + nd \rangle$ below is read off from the constant terms. Table 1 lists every move from a nonnegative element of S_k; the moves from negative elements follow by symmetry.

For instance, from $r = 0$ and $d = 1$ we get $v = 4X + 5 \equiv -1$, so the new carry is $(4X + 6)/3 = 4 \cdot 3^{k-1} + 2 = c - 2 \cdot 3^{k-1}$, using $2X = 6 \cdot 3^{k-1}$. Every carry in the last column lies in S_k, so (ii) holds. Figure 1 draws the result: on the positive side the carries form two ladders that descend

TABLE 1. Moves from the nonnegative carries for $n = 4 \cdot 3^k + 5$ ($X = 3^k$, $c = 2X + 2$, $1 \le i \le k - 1$).

carry r	d	$r + nd$	result
0	$+1$	$4X + 5$	digit -1, carry $c - 2 \cdot 3^{k-1}$
0	-1	$-4X - 5$	digit $+1$, carry $-(c - 2 \cdot 3^{k-1})$
c	$+1$	$6X + 7$	digit $+1$, carry c
c	-1	$-2X - 3$	dead
$c - 2 \cdot 3^i$	$+1$	$6X + 7 - 2 \cdot 3^i$	digit $+1$, carry $c - 2 \cdot 3^{i-1}$
$c - 2 \cdot 3^i$	-1	$-2X - 3 - 2 \cdot 3^i$	dead
$c - 2$	$+1$	$6X + 5$	digit -1, carry c
$c - 2$	-1	$-2X - 5$	digit $+1$, carry $-(c - 4 \cdot 3^{k-1})$
$c - 4 \cdot 3^i$	$+1$	$6X + 7 - 4 \cdot 3^i$	digit $+1$, carry $c - 4 \cdot 3^{i-1}$
$c - 4 \cdot 3^i$	-1	$-2X - 3 - 4 \cdot 3^i$	dead
$c - 4$	$+1$	$6X + 3$	dead
$c - 4$	-1	$-2X - 7$	digit -1, carry $-(c - 4 \cdot 3^{k-1})$

one power of 3 per digit, the bottom of the first ladder feeds c, which repeats forever, and the only other exits lead to the mirror image.

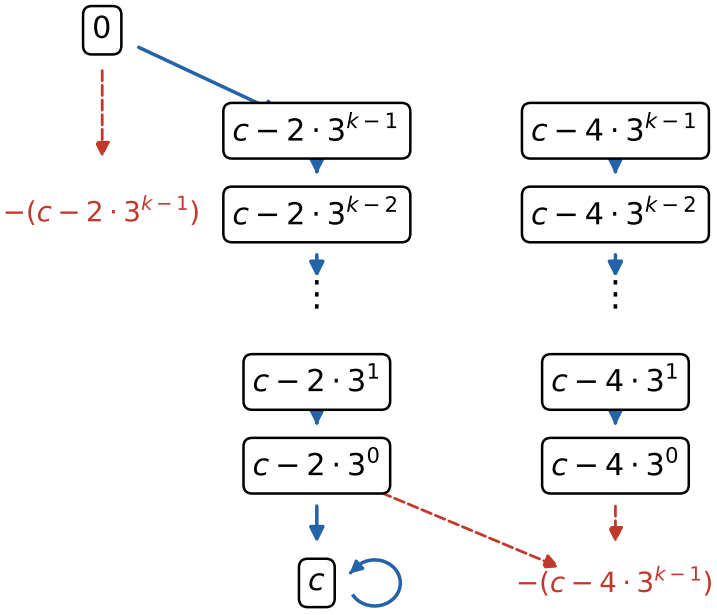


FIGURE 1. The positive half of S_k for $n = 4 \cdot 3^k + 5$, transcribed from Table 1. Solid arrows are moves with $d = 1$, dashed arrows moves with $d = -1$ that are not dead; all other moves are dead. The negative half is the mirror image, and its dashed arrows lead back into this half. Every carry reached by a move with $d = 1$ has a zero digit, so no path ends in an accepting carry.

Bound. Every element of S_k has absolute value at most $c = 2X + 2 < n - 2$, so (iii) holds.

No accepting move. By the table and symmetry, the moves with $d = 1$ that are not dead lead to carries of the form c, $c - 2 \cdot 3^j$ or $c - 4 \cdot 3^j$ with $0 \le j \le k-1$ (the index $j = k-1$ arises from the moves from 0, $-(c-2)$ and $-(c-4)$). All of these are positive. Each has a zero balanced ternary digit, as the following explicit representations show. Write $c = 3^{k+1} - 3^k + 3 - 1$.

- c has nonzero digits only at positions $0, 1, k, k+1$, so its digit at position 2 is 0.
- For $2 \le j \le k-2$, $c - 2 \cdot 3^j = c - 3^{j+1} + 3^j$ and $c - 4 \cdot 3^j = c - 3^{j+1} - 3^j$. The six nonzero digits sit at the distinct positions $0, 1, j, j+1, k, k+1$ among the $k+2$ positions $0, \dots, k+1$, so some digit is 0 when $k \ge 5$.
- $c - 2 = 3^{k+1} - 3^k$ has digit 0 at position 0, and $c - 4 = 3^{k+1} - 3^k - 3 + 1$ has digit 0 at position 2.
- $c - 6 = 3^{k+1} - 3^k - 3 - 1$ has digit 0 at position 2, and $c - 12 = 3^{k+1} - 3^k - 3^2 - 1$ has digit 0 at position 1.
- $c - 2 \cdot 3^{k-1} = 3^k + 3^{k-1} + 3 - 1$ and $c - 4 \cdot 3^{k-1} = 3^k - 3^{k-1} + 3 - 1$ have digit 0 at position 2.

Hence (iv) holds, and Proposition 2.2 gives $4 \cdot 3^k + 5 \notin \mathcal{B}/\mathcal{B}$ for every $k \ge 5$. For $k = 4$ the second item fails at $j = 2$: indeed $329 = 4 \cdot 3^4 + 5$ lies in $\mathcal{B}/\mathcal{B}$, since $329 \cdot 4 = 1316$ and $4 = (11)_3$, $1316 = (1\bar{1}\bar{1}11\bar{1}1\bar{1})_3$ in balanced ternary with $\bar{1} = -1$.

4. The remaining families

For the remaining families the carry sets are larger and depend on the residue of k modulo 2 or 4, but they have the same shape. Write $X = 3^k$. A *chain* is a set

$$\pm\{\, gX + \beta - t \cdot 3^i : i \in I \,\},$$

where g is a positive integer, β and t are integers, and I is either an interval $\{\ell \le i \le k - h\}$ (with $h \ge -1$) or the elements of such an interval in a fixed residue class modulo 2 or 4. A *sporadic pair* is a set $\pm\{gX + s\}$ with g a positive rational whose denominator divides 27 and s an integer. Appendix A lists, for each family and each residue class of k concerned, a set S_k consisting of 0, a few chains and a few sporadic pairs; in every case $S_k = -S_k$ and $|S_k|$ is linear in k. For example, when $n = 4 \cdot 3^k - 5$,

$$S_k = \{0\} \cup \pm\left\{\tfrac{2}{3}X,\ \tfrac{2}{3}X + 2\right\} \cup \pm\{2X - 2 - 2\cdot 3^i\}_{0 \le i \le k-1} \cup \pm\{2X - 2 - 4\cdot 3^i,\ 2X - 2 - 8\cdot 3^i\}_{0 \le i \le k-2}.$$

Condition (i) of Proposition 2.2 holds because every table lists 0. Condition (iii) is a size check: in every table the coefficient g is at most $a/2$, so the elements have absolute value close to $n/2$ at most, well below $n - 2$; the formal proof of Section 5 checks the exact inequality for every family. Conditions (ii) and (iv) reduce to finitely many cases, uniformly in k, through the following two observations.

Lemma 4.1 (chain step)**.** *Let $n = a \cdot 3^k + b$, let $r = \sigma(gX + \beta - t \cdot 3^i)$ with $\sigma = \pm 1$, $i \ge 1$ and $k \ge 1$, and let $d = \pm 1$. The move (r, d) is dead if and only if $\langle \sigma\beta + db \rangle = 0$. Otherwise it produces the digit $e = \langle \sigma\beta + db \rangle$ and leads to*

$$r' = \frac{\sigma g + da}{3} X \ - \ \sigma t \cdot 3^{i-1} \ + \ \frac{\sigma\beta + db - e}{3}.$$

Proof. We have $r + nd = (\sigma g + da)X - \sigma t \cdot 3^i + (\sigma\beta + db)$, and the first two terms are divisible by 3. □

So a move from a chain element of index $i \ge 1$ either dies for every such i or lands on the element of index $i - 1$ of a set of the same form, $\pm\{g'X + \beta' - t' \cdot 3^i\}$, whose parameters are determined by (σ, g, β, t, d) alone (a priori g' is only rational); closure then asks that this set be one of the chains of S_k, or lie in S_k over the relevant indices. Checking (ii) for all but boundedly

many indices therefore amounts to comparing parameters and index ranges. The remaining moves, from the lowest indices of each chain, from the sporadic pairs and from 0, are finitely many explicit computations of the kind in Table 1, whose targets have index $k-j$ or a fixed small index.

Lemma 4.2 (gap)**.** *Let $x = H \cdot 3^s + R$ with integers H, R and $s \geq 1$. Suppose the balanced ternary digit of R at some position $p < s$ is 0 and $|x| > (3^{p+1}-1)/2$. Then x is not zero-free.*

Proof. The balanced digit of x at position p depends only on x modulo 3^{p+1}, and $x \equiv R \pmod{3^{p+1}}$, so it is 0. Since $|x| > (3^{p+1}-1)/2$, the representation of x has more than $p+1$ digits, so this zero is a genuine digit. □

Every positive target of a move with $d = 1$ is either $gX + \beta - t \cdot 3^i$ with i large, to which Lemma 4.2 applies with $s = i$ and $R = \beta$, or $gX + R$ with a fixed integer R (a sporadic element, or a chain element of small index), to which it applies with $s = k - m$, where 3^m is the denominator of g (in our tables $m \leq 2$). In each case p is the position of the first zero digit of R, which is at most the length of R. This gives (iv) for all large k.

For these families the case analysis is too long to print, and Theorems 1.1 and 1.3 rest on a computer verification of it, in Lean 4 as described in the next section: symbolically for $k \geq 14$, and for the finitely many k between the lower bound and 13 through the exact reachable carry set, on which (i) to (iv) are checked directly. For the five families of Theorem 1.1 the symbolic analysis was also carried out independently by a checker in Python that represents carries as linear forms in X and 3^i; it treats k by parity only, so it does not cover the residue classes modulo 4 of Theorem 1.3. The sharpness claims follow from explicit witnesses: with $\bar{1} = -1$,

$$319 \cdot 4 = 1276, \qquad 655 \cdot 2 = 1310, \qquad 665 \cdot 2 = 1330$$

for $4 \cdot 3^4 - 5$, $8 \cdot 3^4 + 7$ and $8 \cdot 3^4 + 17$, where $4 = (11)_3$, $2 = (1\bar{1})_3$ and 1276, 1310, 1330 are zero-free; for $8 \cdot 3^6 - 7 = 5825$ the smallest witness is $m = 8620770364$, with $5825\,m = 50215987370300$ zero-free. (The number $8 \cdot 3^5 - 7 = 1937$ is also an exception; it is in the list of [1], but it is not covered by the uniform argument.)

5. Formal verification

All thirteen non-membership results of Theorems 1.1 and 1.3 are proved in Lean 4 [7] (version 4.34, core library only, without Mathlib). The statement proved for $8 \cdot 3^k + 17$, for example, is

$$\forall k \geq 5, \quad \neg\, \exists\, m, N \in \mathbb{Z}, \;\; 0 < m \;\wedge\; \mathtt{ZF}\, m \;\wedge\; 0 < N \;\wedge\; \mathtt{ZF}\, N \;\wedge\; N = (8 \cdot 3^k + 17)\, m,$$

where `ZF` is defined inductively: 1 and -1 are zero-free, and if q is zero-free then so are $3q \pm 1$. The development has three layers.

(1) A shared file proves Proposition 2.2 for an arbitrary predicate S, and a second version for a concrete finite list of carries, whose conditions are checked by a Boolean function with a proved soundness lemma, so that a concrete instance is closed by evaluation (`decide`).

(2) For each family and each residue class of k, the set S_k is an inductive predicate with one constructor per chain or sporadic pair. Closure is proved constructor by constructor: the lowest indices are split off explicitly, the generic index is sent to $i-1$ as in Lemma 4.1, and every resulting identity or congruence is discharged by the decision procedure `omega` for linear integer arithmetic, with the powers $3^k, 3^{k-1}, \ldots$ and $3^i, 3^{i-1}, \ldots$ as atoms tied together by the equations $3^{x+1} = 3 \cdot 3^x$. The absence of positive zero-free carries is proved by peeling digits as in Lemma 4.2. These lemmas hold for all $k \geq 14$.

(3) The values of k from the lower bound up to 13 are handled by concrete certificates: the exact reachable carry set, checked by evaluation.

The proof scripts for all but the first family were produced by a program that reads the carry sets of Appendix A and decides, for each case, which constructor and index the move lands on. The program's choices are not trusted: Lean re-checks each one, so an incorrect choice makes the file fail to compile. The thirteen main theorems depend only on the standard axioms `propext`, `Quot.sound` and `Classical.choice`; there is no `sorry` and no use of `native_decide`. The whole development (about 31,000 lines, most of them generated) compiles in about three minutes on a laptop. As a check that the verification can fail, we compiled eight corrupted versions: claiming the result for $8 \cdot 3^k + 19$ or for $40 \cdot 3^k + 43$, changing a chain coefficient throughout (in two families), extending $4 \cdot 3^k - 5$ to $k = 4$ or $5 \cdot 3^k + 4$ to $k = 4$, shifting a sporadic carry, and claiming $5 \cdot 3^k + 4 \notin \mathcal{B}/\mathcal{B}$ for $k \equiv 1 \pmod 4$. Each was rejected. A hand-written Lean proof of the case $4 \cdot 3^k + 5$, following Section 3, was also checked separately.

6. Digits 0 and 1

Guy's entry F31 also asks which integers are quotients of numbers written in base 3 with the digits 0 and 1 only. Following [1], let C be this set of positive integers, $V = \mathbb{N} \cap C/C$,

$$D = \{N : N \equiv 2 \cdot 3^{i-1} \pmod{3^i} \text{ for some } i \geq 1\}, \qquad E = \mathbb{N} \cap \bigcup_{i \geq 0} \left[\tfrac{3}{2} \cdot 3^i,\, 2 \cdot 3^i\right],$$

and $F = \mathbb{N} \setminus (V \cup D \cup E)$. Bai et al. show, by looking at numerators and denominators modulo 3^i and at their leading digits, that elements of $D \cup E$ do not lie in V, and they report that Haque and Shallit found the elements

$$529,\ 592,\ 601,\ 616,\ 5368,\ 50281,\ 4072741,\ 4074361,\ 4088941,\ 4245688$$

of F, asking whether F is infinite and suggesting that $621 \cdot 3^{4k} - 20$ might always lie in F. Anders, Dawsey, Reznick and Sisneros-Thiry [2] study the same set and report that these ten are the only elements congruent to 1 modulo 3 below $6.2 \cdot 10^6$, and that Haque and Shallit found eleven below $1.07 \cdot 10^8$. The OEIS entry A339636 [6] lists the elements up to $3.44 \cdot 10^8$ (computed by R. Dougherty-Bliss); it includes $621 \cdot 3^{12} - 20 = 330024841$, the case $k = 3$ of the guess, whose cases $k = 1, 2$ are 50281 and 4074361. (Only elements congruent to 1 modulo 3 need listing, since V, D and E are invariant under $N \mapsto 3N$ and $N \mapsto N/3$, and numbers congruent to 2 modulo 3 lie in D.)

The carry automaton of Section 2, with digits $0, 1$ in place of ± 1, decides membership in V exactly: the proof of Proposition 2.2 and the remark after it carry over word for word, with a move dead when its output digit would be 2 and with acceptance at a carry that is 0 or has only the digits 0 and 1. The carries stay between 0 and $n/2$, so the search is finite. Running it on the next case gives one new data point.

Proposition 6.1. $621 \cdot 3^{16} - 20 = 26732013721$ *lies in* F.

The number of reachable carries for $621 \cdot 3^{4k} - 20$ is 870, 14368, 230658 and 3691740 for $k = 1, \ldots, 4$, growing by a factor close to $16 = 2^4$ per step, that is, like $N^{\log 2/\log 3}$. This Cantor-type growth is why the chain method of Section 4, which relies on carry sets of linear size, does not apply directly to this family, and we do not claim a proof for all k.

7. Data and questions

A search with the decision procedure of Section 2 finds exactly 200 integers $n \not\equiv 0 \pmod 3$ below 10^8 outside $\mathcal{B}/\mathcal{B}$. Of these, 136 fall into 21 families $a \cdot 3^k + b$ with at least three members each, listed in Table 2; the other 64 belong to no family with $a, |b| \leq 300$ and at least three members below 10^8. Here a family is written in normalised form, with $3 \nmid a$ and $|b| < 3^{k-1}$ for every member counted, so that each exception has at most one representation $a \cdot 3^k + b$ for a

given k and families are not repeated under $a \mapsto 3a$, $k \mapsto k-1$. Without the normalisation the same 136 exceptions are spread over 77 parameter pairs. Within its range, each family is periodic in k from some point on, with period 1, 2 or 4. The families also come in visible pairs: b changes sign, as with $4 \cdot 3^k \pm 5$, or the roles of a and b swap, as with $40 \cdot 3^k \pm 41$ and $41 \cdot 3^k \pm 40$. Theorems 1.1 and 1.3 account for every exception in ten of the families within this range, apart from $1937 = 8 \cdot 3^5 - 7$, which is known only by computation, and for three more families on one residue class. For the remaining eight, and for $5 \cdot 3^k \pm 4$ on the other residue classes, we found no carry set of linear size. For $5 \cdot 3^k \pm 4$ the reachable carry sets grow roughly quadratically in k, which suggests elements with two independent indices. For $3^k \pm 4$, $7 \cdot 3^k \pm 8$ and $19 \cdot 3^k - 10$ they grow by a factor of about 15 each time k increases by 4, like the Cantor-type growth of Section 6. For $40 \cdot 3^k - 41$ and $41 \cdot 3^k \pm 40$ they have thousands of elements already for k near 13. Proving these families needs a different description of the carries.

TABLE 2. Families $a \cdot 3^k + b$ among the 200 exceptions below 10^8. The pattern shows $k = 1, 2, \ldots$ up to $a \cdot 3^k + b < 10^8$: x marks an exception, a dot a member of $\mathcal{B}/\mathcal{B}$.

family	pattern	
$4 \cdot 3^k - 5$	`....xxxxxxxxxx`	proved
$4 \cdot 3^k + 5$	`....xxxxxxxxxx`	proved
$5 \cdot 3^k - 4$	`....xxxxxxxxxx`	proved for $k \equiv 2$ (4)
$5 \cdot 3^k + 4$	`....xxxxxxxxxx`	proved for $k \equiv 0$ (4)
$8 \cdot 3^k + 7$	`....xxxxxxxxx`	proved
$8 \cdot 3^k + 17$	`....xxxxxxxxx`	proved
$8 \cdot 3^k - 7$	`....x.xxxxxxx`	proved for $k \geq 7$
$3^k + 4$	`....x.x.x.x.x.x.`	
$3^k - 4$	`.....x.x.x.x.x.`	
$7 \cdot 3^k - 8$	`.....x.x.x.x.x`	
$7 \cdot 3^k + 8$	`.....x.x.x.x.x`	
$8 \cdot 3^k - 17$	`.....x.x.x.x.x`	proved
$16 \cdot 3^k + 17$	`.....x.x.x.x.x`	proved
$40 \cdot 3^k - 41$	`........xxxxx`	
$40 \cdot 3^k + 41$	`........xxxxx`	proved for $k \equiv 0$ (4)
$41 \cdot 3^k - 40$	`........xxxxx`	
$41 \cdot 3^k + 40$	`........xxxxx`	
$10 \cdot 3^k - 19$	`.....x...x...x`	proved
$10 \cdot 3^k - 11$	`.....x...x...x`	proved
$19 \cdot 3^k - 10$	`.....x...x...x`	
$20 \cdot 3^k - 19$	`.....x...x...x`	proved

Question 7.1. Is $5 \cdot 3^k + 4 \notin \mathcal{B}/\mathcal{B}$ for all $k \geq 5$? Is $3^k + 4 \notin \mathcal{B}/\mathcal{B}$ for all odd $k \geq 5$? (The second question is Shallit's: he suggests these numbers as counterexamples in the OEIS entry A351243 [6], 5 February 2022.)

Question 7.2. Does $\mathcal{B}/\mathcal{B}$ contain almost all integers prime to 3, that is, do the exceptions have density zero?

Question 7.3. Is F infinite? In particular, is $621 \cdot 3^{4k} - 20 \in F$ for every $k \geq 1$?

Code and data

The Lean development, the program that generates it, the decision procedures, the search program that finds the 200 exceptions below 10^8 (about 25 minutes), the list it produces, and a program that recomputes the numbers quoted in this paper are at

https://github.com/jvvk/mathematics/tree/main/balanced-ternary-quotients.

The recomputation program checks that each listed number is outside $\mathcal{B}/\mathcal{B}$, the family counts, the witnesses and the carry sets of Appendix A against the criterion; completeness of the list below 10^8 rests on the search program, which is run separately.

Changes in version 3. The second part of Question 7.1 is credited to Shallit; the statements of what Lean verifies and what the programs check are narrowed to what they do; the normalisation behind Table 2 is stated; a sign in the upper-gap test of the independent symbolic checker is corrected, and its certificates for the five families of Theorem 1.1 were rerun and pass. The theorems, their proofs and the Lean files are unchanged.

Declaration of generative AI use

The author used Claude (Anthropic; Claude Opus models, between 19 September and 8 October 2026) throughout this work, under the author's direction: to run the searches for exceptions and fit the carry sets of Appendix A, to write the decision programs, the symbolic checker, the program that generates the Lean proofs and the hand-written Lean proof, to search the literature, and to draft the text. The reason was to obtain a computer-assisted proof that can be checked in full: the thirteen family non-membership statements of Theorems 1.1 and 1.3 are verified in Lean (Section 5), and the witnesses and the other finite claims are recomputed by a separate program (see *Code and data*). The author takes full responsibility for the content, including the references.

References

[1] J. H. Bai, J. Meleshko, S. Riasat and J. Shallit, Quotients of palindromic and antipalindromic numbers, *Integers* **22** (2022), Paper A96, https://math.colgate.edu/~integers/w96/w96.pdf; arXiv:2202.13694.

[2] K. Anders, M. L. Dawsey, B. Reznick and S. Sisneros-Thiry, Representations of integers as quotients of sums of distinct powers of three, arXiv:2308.07252, 2023.

[3] R. K. Guy, *Unsolved Problems in Number Theory*, 3rd edition, Problem Books in Mathematics, Springer, New York, 2004, doi:10.1007/978-0-387-26677-0.

[4] J. H. Loxton and A. J. van der Poorten, An awful problem about integers in base four, *Acta Arith.* **49** (1987), no. 2, 193–203, doi:10.4064/aa-49-2-193-203.

[5] E. Rowland and J. Shallit, Automatic sets of rational numbers, *Internat. J. Found. Comput. Sci.* **26** (2015), no. 3, 343–365, doi:10.1142/S0129054115500197.

[6] OEIS Foundation Inc., The On-Line Encyclopedia of Integer Sequences, https://oeis.org, entries A339636 and A351243, accessed 8 October 2026.

[7] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction, CADE 28*, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635, doi:10.1007/978-3-030-79876-5_37.

Appendix A. Carry sets

In each table $X = 3^k$, and every row stands for the listed element together with its negative. The sizes $|S_k|$ are for $k \geq 14$ in the stated parity class.

TABLE 3. Carry set S_k for $n = 4 \cdot 3^k - 5$, all k; $X = 3^k$, $|S_k| = 6k + 1$.

element	index range
0	
$\pm\big(2X - 2 - 8 \cdot 3^i\big)$	$0 \le i \le k-2$
$\pm\big(2X - 2 - 2 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(2X - 2 - 4 \cdot 3^i\big)$	$0 \le i \le k-2$
$\pm\big(\frac{2}{3}X\big)$	sporadic
$\pm\big(\frac{2}{3}X + 2\big)$	sporadic

TABLE 4. Carry set S_k for $n = 8 \cdot 3^k + 7$, k even; $X = 3^k$, $|S_k| = 4k + 3$.

element	index range
0	
$\pm\big(2X + 2 - 2 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 2$
$\pm\big(2X + 2 - 4 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 2$
$\pm\big(2X + 2 + 4 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 2$
$\pm\big(2X + 2 + 2 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 2$
$\pm\big(2X + 2\big)$	sporadic

TABLE 5. Carry set S_k for $n = 8 \cdot 3^k + 7$, k odd; $X = 3^k$, $|S_k| = 8k + 3$.

element	index range
0	
$\pm\big(2X + 2 - 2 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(2X + 2 + 4 \cdot 3^i\big)$	$0 \le i \le k-1$, $i \equiv 0 \pmod 2$
$\pm\big(2X + 2 - 4 \cdot 3^i\big)$	$1 \le i \le k-2$, $i \equiv 1 \pmod 2$
$\pm\big(2X + 2 + 2 \cdot 3^i\big)$	$0 \le i \le k$
$\pm\big(4X + 4 - 2 \cdot 3^i\big)$	$1 \le i \le k-1$
$\pm\big(2X + 2\big)$	sporadic

TABLE 6. Carry set S_k for $n = 8 \cdot 3^k + 17$, k even; $X = 3^k$, $|S_k| = 10k + 3$.

element	index range
0	
$\pm\big(2X + 4 - 2 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(2X + 4 - 4 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 2$
$\pm\big(2X + 4 + 4 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 2$
$\pm\big(2X + 4 + 2 \cdot 3^i\big)$	$0 \le i \le k$, $i \equiv 0 \pmod 2$
$\pm\big(2X + 4 + 2 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 2$
$\pm\big(4X + 8 - 2 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(4X + 8 - 4 \cdot 3^i\big)$	$1 \le i \le k-2$
$\pm\big(2X + 4\big)$	sporadic
$\pm\big(\frac{8}{3}X + 6\big)$	sporadic
$\pm\big(\frac{10}{3}X + 6\big)$	sporadic
$\pm\big(4X + 8\big)$	sporadic

TABLE 7. Carry set S_k for $n = 8 \cdot 3^k + 17$, k odd; $X = 3^k$, $|S_k| = 10k + 3$.

element	index range
0	
$\pm\big(2X + 4 - 2 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(2X + 4 + 4 \cdot 3^i\big)$	$0 \le i \le k-3$, $i \equiv 0 \pmod 2$
$\pm\big(2X + 4 - 4 \cdot 3^i\big)$	$1 \le i \le k-2$, $i \equiv 1 \pmod 2$
$\pm\big(2X + 4 + 2 \cdot 3^i\big)$	$0 \le i \le k-3$, $i \equiv 0 \pmod 2$
$\pm\big(2X + 4 + 2 \cdot 3^i\big)$	$1 \le i \le k$, $i \equiv 1 \pmod 2$
$\pm\big(4X + 8 - 2 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(4X + 8 - 4 \cdot 3^i\big)$	$1 \le i \le k-2$
$\pm\big(2X + 4\big)$	sporadic
$\pm\big(\frac{8}{3}X + 6\big)$	sporadic
$\pm\big(\frac{10}{3}X + 6\big)$	sporadic
$\pm\big(4X + 8\big)$	sporadic

TABLE 8. Carry set S_k for $n = 8 \cdot 3^k - 7$, k even; $X = 3^k$, $|S_k| = 12k + 5$.

element	index range
0	
$\pm\big(2X - 2 - 2 \cdot 3^i\big)$	$0 \le i \le k-2$
$\pm\big(2X - 2 - 4 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 2$
$\pm\big(2X - 2 - 10 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 2$
$\pm\big(2X - 2 + 16 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 2$
$\pm\big(2X - 2 - 16 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 2$
$\pm\big(2X - 2 + 10 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 2$
$\pm\big(2X - 2 + 4 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 2$
$\pm\big(2X - 2 + 2 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(4X - 4 - 2 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(\frac{4}{3}X\big)$	sporadic
$\pm\big(2X - 2\big)$	sporadic
$\pm\big(\frac{10}{3}X\big)$	sporadic
$\pm\big(\frac{10}{3}X + 2\big)$	sporadic
$\pm\big(4X - 4\big)$	sporadic

TABLE 9. Carry set S_k for $n = 8 \cdot 3^k - 7$, k odd; $X = 3^k$, $|S_k| = 4k + 3$.

element	index range
0	
$\pm\big(2X - 2 - 2 \cdot 3^i\big)$	$1 \le i \le k-2$, $i \equiv 1 \pmod 2$
$\pm\big(2X - 2 + 4 \cdot 3^i\big)$	$0 \le i \le k-1$, $i \equiv 0 \pmod 2$
$\pm\big(2X - 2 - 4 \cdot 3^i\big)$	$1 \le i \le k-2$, $i \equiv 1 \pmod 2$
$\pm\big(2X - 2 + 2 \cdot 3^i\big)$	$0 \le i \le k-1$, $i \equiv 0 \pmod 2$
$\pm\big(2X - 2\big)$	sporadic

TABLE 10. Carry set S_k for $n = 8 \cdot 3^k - 17$, $k \equiv 0 \pmod 2$; $X = 3^k$, $|S_k| = 16k - 3$.

element	index range
0	
$\pm\big(4X - 8 - 8 \cdot 3^i\big)$	$0 \le i \le k-3$
$\pm\big(4X - 8 - 16 \cdot 3^i\big)$	$0 \le i \le k-3$
$\pm\big(4X - 8 - 2 \cdot 3^i\big)$	$0 \le i \le k$
$\pm\big(4X - 8 - 4 \cdot 3^i\big)$	$0 \le i \le k-2$
$\pm\big(2X - 4 - 2 \cdot 3^i\big)$	$0 \le i \le k-2$
$\pm\big(2X - 4 - 4 \cdot 3^i\big)$	$2 \le i \le k-2$, $i \equiv 0 \pmod 2$
$\pm\big(2X - 4 - 10 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 2$
$\pm\big(2X - 4 + 10 \cdot 3^i\big)$	$0 \le i \le k-4$, $i \equiv 0 \pmod 2$
$\pm\big(2X - 4 + 4 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 2$
$\pm\big(2X - 4 + 2 \cdot 3^i\big)$	$0 \le i \le k-3$
$\pm\big(\frac{4}{3}X - 2\big)$	sporadic
$\pm\big(\frac{4}{3}X\big)$	sporadic
$\pm\big(\frac{4}{3}X + 2\big)$	sporadic
$\pm\big(2X - 4\big)$	sporadic
$\pm\big(\frac{20}{9}X - 6\big)$	sporadic
$\pm\big(\frac{8}{3}X - 6\big)$	sporadic
$\pm\big(\frac{28}{9}X - 6\big)$	sporadic
$\pm\big(\frac{10}{3}X - 6\big)$	sporadic

TABLE 11. Carry set S_k for $n = 16 \cdot 3^k + 17$, $k \equiv 0 \pmod 2$; $X = 3^k$, $|S_k| = 12k + 11$.

element	index range
0	
$\pm\big(4X + 4 - 4 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(4X + 4 - 8 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 2$
$\pm\big(4X + 4 + 8 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 2$
$\pm\big(4X + 4 + 4 \cdot 3^i\big)$	$0 \le i \le k$, $i \equiv 0 \pmod 2$
$\pm\big(4X + 4 + 4 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 2$
$\pm\big(8X + 8 - 8 \cdot 3^i\big)$	$0 \le i \le k-2$
$\pm\big(8X + 8 - 16 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(8X + 8 - 4 \cdot 3^i\big)$	$1 \le i \le k-1$
$\pm\big(\frac{8}{3}X + 6\big)$	sporadic
$\pm\big(4X + 4\big)$	sporadic
$\pm\big(4X + 6\big)$	sporadic
$\pm\big(\frac{16}{3}X + 6\big)$	sporadic
$\pm\big(\frac{20}{3}X + 6\big)$	sporadic
$\pm\big(8X + 6\big)$	sporadic
$\pm\big(8X + 8\big)$	sporadic

TABLE 12. Carry set S_k for $n = 10 \cdot 3^k - 11$, $k \equiv 2 \pmod 4$; $X = 3^k$, $|S_k| = 4k + 5$.

element	index range
0	
$\pm\big(2X - 2 - 2 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(2X - 2 + 4 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(2X - 2 - 4 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(2X - 2 + 2 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(4X - 4 - 2 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(4X - 4 - 4 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(4X - 4 + 4 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(4X - 4 + 2 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(2X - 2\big)$	sporadic
$\pm\big(4X - 4\big)$	sporadic

TABLE 13. Carry set S_k for $n = 10 \cdot 3^k - 19$, $k \equiv 2 \pmod 4$; $X = 3^k$, $|S_k| = 10k + 3$.

element	index range
0	
$\pm\big(2X - 4 - 2 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(2X - 4 - 2 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(2X - 4 - 2 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(2X - 4 + 4 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(2X - 4 + 4 \cdot 3^i\big)$	$1 \le i \le k-5$, $i \equiv 1 \pmod 4$
$\pm\big(2X - 4 - 4 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(2X - 4 - 4 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(2X - 4 + 2 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(2X - 4 + 2 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(2X - 4 + 2 \cdot 3^i\big)$	$2 \le i \le k$, $i \equiv 2 \pmod 4$
$\pm\big(4X - 8 - 2 \cdot 3^i\big)$	$1 \le i \le k-5$, $i \equiv 1 \pmod 4$
$\pm\big(4X - 8 - 2 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(4X - 8 - 2 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(4X - 8 - 4 \cdot 3^i\big)$	$1 \le i \le k-5$, $i \equiv 1 \pmod 4$
$\pm\big(4X - 8 - 4 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(4X - 8 + 4 \cdot 3^i\big)$	$4 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(4X - 8 + 4 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(4X - 8 + 2 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(4X - 8 + 2 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(4X - 8 + 2 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(2X - 4\big)$	sporadic
$\pm\big(\frac{8}{3}X - 6\big)$	sporadic
$\pm\big(\frac{10}{3}X - 6\big)$	sporadic
$\pm\big(4X - 8\big)$	sporadic

TABLE 14. Carry set S_k for $n = 20 \cdot 3^k - 19$, $k \equiv 2 \pmod 4$; $X = 3^k$, $|S_k| = 12k + 11$.

element	index range
0	
$\pm\big(4X - 4 - 4 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(4X - 4 - 4 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(4X - 4 - 4 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(4X - 4 - 8 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(4X - 4 - 8 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(4X - 4 + 16 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(4X - 4 - 16 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(4X - 4 + 8 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(4X - 4 + 8 \cdot 3^i\big)$	$1 \le i \le k-5$, $i \equiv 1 \pmod 4$
$\pm\big(4X - 4 + 4 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(4X - 4 + 4 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(4X - 4 + 4 \cdot 3^i\big)$	$2 \le i \le k$, $i \equiv 2 \pmod 4$
$\pm\big(8X - 8 - 4 \cdot 3^i\big)$	$1 \le i \le k-5$, $i \equiv 1 \pmod 4$
$\pm\big(8X - 8 - 4 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(8X - 8 - 4 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(8X - 8 - 8 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(8X - 8 - 8 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(8X - 8 + 16 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(8X - 8 - 16 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(8X - 8 + 8 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(8X - 8 + 8 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(8X - 8 + 4 \cdot 3^i\big)$	$4 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(8X - 8 + 4 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(8X - 8 + 4 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(4X - 6\big)$	sporadic
$\pm\big(4X - 4\big)$	sporadic
$\pm\big(\frac{16}{3}X - 6\big)$	sporadic
$\pm\big(\frac{20}{3}X - 6\big)$	sporadic
$\pm\big(8X - 8\big)$	sporadic
$\pm\big(8X - 6\big)$	sporadic
$\pm\big(\frac{28}{3}X - 6\big)$	sporadic

TABLE 15. Carry set S_k for $n = 5 \cdot 3^k - 4$, $k \equiv 2 \pmod 4$; $X = 3^k$, $|S_k| = 4k + 5$.

element	index range
0	
$\pm\big(X - 1 - 2 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(X - 1 + 4 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(X - 1 - 4 \cdot 3^i\big)$	$3 \le i \le k-3$, $i \equiv 3 \pmod 4$
$\pm\big(X - 1 + 2 \cdot 3^i\big)$	$1 \le i \le k-1$, $i \equiv 1 \pmod 4$
$\pm\big(2X - 2 - 2 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(2X - 2 + 4 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(2X - 2 - 4 \cdot 3^i\big)$	$2 \le i \le k-4$, $i \equiv 2 \pmod 4$
$\pm\big(2X - 2 + 2 \cdot 3^i\big)$	$0 \le i \le k-2$, $i \equiv 0 \pmod 4$
$\pm\big(X - 1\big)$	sporadic
$\pm\big(2X - 2\big)$	sporadic

TABLE 16. Carry set S_k for $n = 5 \cdot 3^k + 4$, $k \equiv 0 \pmod 4$; $X = 3^k$, $|S_k| = 4k + 5$.

element	index range
0	
$\pm\big(X + 1 - 2 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 4$
$\pm\big(X + 1 + 4 \cdot 3^i\big)$	$3 \le i \le k-1$, $i \equiv 3 \pmod 4$
$\pm\big(X + 1 - 4 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 4$
$\pm\big(X + 1 + 2 \cdot 3^i\big)$	$3 \le i \le k-1$, $i \equiv 3 \pmod 4$
$\pm\big(2X + 2 - 2 \cdot 3^i\big)$	$0 \le i \le k-4$, $i \equiv 0 \pmod 4$
$\pm\big(2X + 2 - 4 \cdot 3^i\big)$	$0 \le i \le k-4$, $i \equiv 0 \pmod 4$
$\pm\big(2X + 2 + 4 \cdot 3^i\big)$	$2 \le i \le k-2$, $i \equiv 2 \pmod 4$
$\pm\big(2X + 2 + 2 \cdot 3^i\big)$	$2 \le i \le k-2$, $i \equiv 2 \pmod 4$
$\pm\big(X + 1\big)$	sporadic
$\pm\big(2X + 2\big)$	sporadic

TABLE 17. Carry set S_k for $n = 40 \cdot 3^k + 41$, $k \equiv 0 \pmod 4$; $X = 3^k$, $|S_k| = 12k + 19$.

element	index range
0	
$\pm\big(20X + 20 - 40 \cdot 3^i\big)$	$0 \le i \le k-1$
$\pm\big(20X + 20 - 20 \cdot 3^i\big)$	$0 \le i \le k-3$
$\pm\big(20X + 20 - 4 \cdot 3^i\big)$	$0 \le i \le k+1$
$\pm\big(8X + 8 + 32 \cdot 3^i\big)$	$3 \le i \le k-1$, $i \equiv 3 \pmod 4$
$\pm\big(8X + 8 - 32 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 4$
$\pm\big(8X + 8 - 4 \cdot 3^i\big)$	$3 \le i \le k-1$, $i \equiv 3 \pmod 4$
$\pm\big(8X + 8 + 16 \cdot 3^i\big)$	$3 \le i \le k-5$, $i \equiv 3 \pmod 4$
$\pm\big(8X + 8 - 16 \cdot 3^i\big)$	$1 \le i \le k-3$, $i \equiv 1 \pmod 4$
$\pm\big(8X + 8 + 4 \cdot 3^i\big)$	$5 \le i \le k-3$, $i \equiv 1 \pmod 4$
$\pm\big(16X + 16 - 32 \cdot 3^i\big)$	$0 \le i \le k-4$, $i \equiv 0 \pmod 4$
$\pm\big(16X + 16 + 32 \cdot 3^i\big)$	$2 \le i \le k-2$, $i \equiv 2 \pmod 4$
$\pm\big(16X + 16 - 4 \cdot 3^i\big)$	$2 \le i \le k-2$, $i \equiv 2 \pmod 4$
$\pm\big(16X + 16 - 16 \cdot 3^i\big)$	$0 \le i \le k-4$, $i \equiv 0 \pmod 4$
$\pm\big(16X + 16 + 16 \cdot 3^i\big)$	$2 \le i \le k-6$, $i \equiv 2 \pmod 4$
$\pm\big(16X + 16 + 4 \cdot 3^i\big)$	$4 \le i \le k-4$, $i \equiv 0 \pmod 4$
$\pm\big(\frac{20}{3}X + 14\big)$	sporadic
$\pm\big(8X + 8\big)$	sporadic
$\pm\big(8X + 14\big)$	sporadic
$\pm\big(\frac{40}{3}X + 14\big)$	sporadic
$\pm\big(\frac{140}{9}X + 18\big)$	sporadic
$\pm\big(16X + 16\big)$	sporadic
$\pm\big(16X + 18\big)$	sporadic
$\pm\big(\frac{160}{9}X + 18\big)$	sporadic
$\pm\big(\frac{56}{3}X + 14\big)$	sporadic
$\pm\big(\frac{176}{9}X + 18\big)$	sporadic
$\pm\big(20X + 14\big)$	sporadic
$\pm\big(20X + 18\big)$	sporadic
$\pm\big(20X + 20\big)$	sporadic